

Solution Architecture Specification

Date	06 July 2026
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Team

Solution Architecture Layers

Layer	Description / Role in Architecture	Technologies Used
Presentation Layer	Responsive dashboard layout rendering static pages, form inputs, batch upload drop zones, and dynamic evaluation badges.	HTML5, CSS3, Vanilla JS, Font Awesome
API Gateway / Routing	Manages web requests, routing static traffic, JSON single scores, and multipart CSV batch file streams.	Flask HTTP server framework
Core Logic Service	Preprocesses inputs, loads pickled scaler, maps fields, runs XGBoost model classifications, and checks risk flags.	Python, scikit-learn, XGBoost, pandas, numpy
Data / Storage Layer	Manages training records, outputs generated batch CSV, and loads pickled model files.	CSV, Pickled serialization (rdf.pkl)